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Technical Reproducibility Validation Report

STRait Razor

Version b618e93
STR / SNP panel

Verification date	2026-08-14
Repository commit	b618e9345ab40f348b504083ae8de2b39abb60fa
Attestation level	**Runs + Plausible output**
Permalink	https://strhub.app/verified/strait-razor-b618e93
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**5/5 gates passed. Runs + Plausible output**
ATTESTATION LEVEL	**Runs + Plausible output**
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	STRait Razor
Version	b618e93
Repository	https://github.com/Ahhgust/STRaitRazor
Commit (pinned)	b618e9345ab40f348b504083ae8de2b39abb60fa
Container OS	ubuntu-22.04
Marker panel	STR / SNP panel
Verified on	2026-08-14 12:30:37 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/31800572221
Submitted by	A third party (not the tool's maintainer)

This tool was submitted for verification by somebody other than its maintainer. The maintainer took no part in the run and supplied none of what it used: the command, the environment, and any target regions were chosen by the submitter. This certificate records a run; it is not an endorsement by the tool's maintainer, and does not imply their involvement.

3. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · STRait Razor b618e93 · commit b618e93
$ /opt/tool/str8rzr \
-c /opt/tool/ForenSeqv1.27.config \
/data/in/sample.fastq > /data/out/allsequences.txt
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	PASS
Execution log	—	

5. Out of Scope

This report does not evaluate any of the following:

– Genotype correctness or accuracy
– Concordance against known truth sets
– Sensitivity, specificity, or stutter performance
– Allele calling accuracy or forensic casework suitability
– Regulatory compliance or ISO accreditation
– Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	allsequences.txt
Format	TSV
Sequence records	816
Distinct STR loci	63
Total reads	3,967
Max depth (single locus)	132

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01)
Source	https://data.nist.gov/od/id/mds2-2157
License	research / training / education only (per NIST); not for donor identification or database searching
Ref. genome	—

Loci tested	Amelogenin · CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D17S1301 · D18S51 · D19S433 · D1S1656 · D20S482 · D21S11 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D4S2408 · D5S818 · D6S1043 · D7S820 · D8S1179 · D9S1122 · DXS10074 · DXS10103 · DXS10135 · DXS7132 · DXS7423 · DXS8378 · DYF387S1 · DYS19 · DYS385 · DYS389I · DYS389II · DYS390 · DYS391 · DYS392 · DYS437 · DYS438 · DYS439 · DYS448 · DYS460 · DYS461 · DYS481 · DYS505 · DYS522 · DYS533 · DYS549 · DYS570 · DYS576 · DYS612 · DYS635 · DYS643 · FGA · HPRTB · N29insA · PentaD · PentaE · TH01 · TPOX · Y-GATA-H4 · mh16-MC1RB · mh16-MC1RC · vWA

8. Verification Matrix

PASS	Reference dataset	NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.

10. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

11. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

12. Conclusion

◆ **Runs end-to-end**

STRait Razor b618e93 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 5 verification gates were passed.

◆ **Produces valid output**

The tool generated a structurally valid TSV output file with genotype calls across 63 target forensic STR loci, with a maximum read depth of 132 at DYS392.

◆ **No bundled demo data**

STRait Razor does not include its own test or demo dataset. STRhub Verified supplied NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix: Detected Markers with Read Depth

STRait Razor b618e93 · verified 2026-08-14 · showing the 60 deepest of 220 detected markers by read depth

LOCUS	READ DEPTH	LOCUS	READ DEPTH
DYS392	170	DYS643	37
DYS438	117	HPRTB	36
D6S1043	111	rs4364205	36
TH01	106	D8S1179	34
DYS389I	97	DXS10135	34
DYS576	95	DYF387S1	34
DYS505	93	DYS390	34
D20S482	84	D12S391	31
FGA	81	DYS391	31
D3S1358	69	D21S11	30
D2S441	66	rs1919550	29
DYS612	63	D13S317	28
D16S539	61	rs174570	28
D19S433	59	rs192655	28
DXS10074	58	D17S1301	26
D4S2408	52	CSF1PO	24
DXS7132	51	rs1454361	24
D18S51	48	rs8078417	24
rs200354	47	DYS389II	23
DYS385	45	DYS439	23
D22S1045	44	DYS570	23
DXS8378	44	rs1382387	23
DXS7423	43	rs722098	23
rs1109037	43	PentaD	22
DYS437	40	rs2814778	22
mh16-MC1RC	40	rs876724	22
DYS448	39	rs2402130	21
D10S1248	38	rs4918664	21
rs459920	38	rs917118	21
D7S820	37	DYS481	20